

7. Capture video from web Camera and Display the video, in slow motion and in fast motion.

Aim:

To capture video from a webcam and display it in slow and fast motion using the OpenCV library.

Procedure:

- Import the required libraries such as OpenCV (cv2) and NumPy.
- Read the input image using the cv2.imread() function.
- Create the required kernel/parameters for image processing.
- Apply the required image-processing operation using the appropriate OpenCV function.
- Display the original and processed images using cv2.imshow() and close the windows using cv2.destroyAllWindows().

Code:

```
import cv2

# Load the video

video =

cv2.VideoCapture("/Users/michellejoanna/Downloads/WhatsApp
Video 2026-07-24 at 01.41.14.mp4")

if not video.isOpened():

    print("Cannot open video")

    exit()
```

```
print("===== Video Speed =====")
```

```
print("1. Normal Speed")
```

```
print("2. Fast Motion")
```

```
print("3. Slow Motion")
```

```
choice = input("Enter your choice (1/2/3): ")
```

```
if choice == "1":
```

```
    delay = 30
```

```
    title = "Normal Speed Video"
```

```
elif choice == "2":
```

```
    delay = 5
```

```
    title = "Fast Motion Video"
```

```
elif choice == "3":
```

```
    delay = 100
```

```
    title = "Slow Motion Video"
```

```
else:
```

```
    print("Invalid Choice!")
```

```
    video.release()
```

```
    exit()
```

```
while True:

    ret, frame = video.read()

    if not ret:

        break

    cv2.imshow(title, frame)

    if cv2.waitKey(delay) & 0xFF == ord('q'):

        break

video.release()

cv2.destroyAllWindows()
```

Sample Input

Original Video from the webcam

Sample Output

Video in slow motion and fast motion from webcam

Result:

The video was successfully captured from the webcam and displayed in slow-motion and fast-motion using OpenCV.